Clean Code

The Main Idea: the main idea of Clean Code is not to write a code only for the computer to understand, it should also be clear for programmers who will read or develop it later.

Whit It Is? clean Code is a way of writing the code that is clear and easy to read, understandable by any programmer, modifiable, simple, and well organized.

Clean code usually has:
-Meaningful variable and function names
-Logical project structure
-Less code duplication
-Small functions with one responsibility
-Avoiding unnecessary complexity
-Easier testing and maintenance

Why It exists? it exists to reduce errors, improve teamwork, and make the code easier to update and maintain, reduced development and maintain cost.

Examples:

Bad:

```
String s (String fn, String ln){
return fn+ln; }
```

Good:

```
String getUserName(String firstName, String lastName){
return firstName + lastName; }
```

-Project Structure:

Whit It Is? Project Structure is the way files and folders are arranged inside a software project and it depends on project.

Why It exists?

-**Easy file access:** so developers can quickly find the files they need.
-**Better teamwork**: team members know where each feature's files are.
-**Scalability**: organized projects will be easier to gather with new features.
-**Easier maintenance**: it helps developers fix bugs and update the project faster.
-**Simpler GitHub merging:** a well-structured project makes merging code in GitHub easier and reduces merge conflicts between developers.

How files and Folders Are Organized Based On Project's Idea? choose the project structure based on the **main features** and **complexity** of the project.

-first understand the purpose of the project, its requirements, and the main features it contains. After that, organize the files and folders based on those features.

= Before creating the structure, ask questions like:

-What does the project do?
-What are the main modules or features?
-Is the project small or large?
-Will multiple developers work on it?
-Will new features be added in the future?

Example:

Bad Structure:

```
lib/
├── login.dart
├── register.dart
├── product.dart
├── cart.dart
├── colors.dart
├── profile.dart
├── home.dart
├── payment.dart
└── main.dart
```

Good structure:

```
lib/
├── core/
│   ├── constants/
│   └── services/
├── features/
│   ├── auth/
│   │   ├── models/
│   │   ├── controllers/
│   │   └── screens/
│   ├── cart/
│   │   ├── controllers/
│   │   └── screens/
│   └── profile/
│   ├── models/
│   ├── screens/
│   └── widgets/
│
├── shared/
│   ├── widgets/
│   └── components/
│
└── main.dart
```

-Namings:

Whit It Is? naming is the choosing of clear and meaningful names for variables, functions, files, folders, and the name should describe what the variable, function, file, or folder actually does.

Why It exists? its helps developers understand the purpose of the code without needing extra explanation, so that improves code readability and teamwork, speeds development processes.

Example:

Bad Namings:

```
int x;
String data;

function doStuff();

folder1/
test.dart
test2.dart
```

Good Namings:

```
String userName;
int productId;

function loginUser();

auth/
product_screen.dart
```

-DRY-Don’t Repeat Yourself

Whit It Is? It's means that developers should avoid writing the same code multiple times, instead of repeating code, we can create one code and reuse it.

This will make the code cleaner, shorter, easier to maintain and update.

Why It exists?

-**Reduces code duplication:** repeated code makes projects harder to manage.
-**Makes maintenance easier:** if code exists in one place, updating it becomes simpler.
-**Reduces bugs**: repeating the same logic in different places may create inconsistent behavior and errors.
- **Improves readability:** cleaner and reusable code is easier to understand.
- **Saves development time:** developers can reuse existing code instead of rewriting it.

Example:

Bad

```
print("Welcome Hanaa");
print("Welcome Shahed");
print("Welcome Sara");
```

Good

```
function welcomeUser(name) {
print("Welcome " + name);}

welcomeUser("Hanaa ");
welcomeUser("Shahed");
welcomeUser("Sara");
```

Clean Code, good project structure, clear naming, and the DRY principle are essential practices in software development. When developers follow these rules, projects become more organized, scalable, and less personal error, also they will improve teamwork and reducing confusion during collaboration.
writing good code is not only about making it work, but about making it simple, clear, and professional so it can be easily updated and understood in the future.

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